GinCol Lab · Aula

Student worksheet

Biology, Geology and Environmental Sciences · 1st year of Bachillerato · Inside the Earth · Class 5 of 5 · With Nil

Two ways to slice Earth

Class video: https://youtu.be/BqiwBYVsN3Q

The code leads to the video.

Name: _____ Date: _____

The video is in Spanish, with English subtitles. On the board, *corteza* is crust, *manto* is mantle, *núcleo* is core (*externo*, outer; *interno*, inner), *litosfera* is lithosphere, *astenosfera* is asthenosphere and *temperatura de fusión* is melting temperature. In Spain the decimal separator is a comma: this sheet uses the decimal point.

Before watching the video

1. Draw the inside of the Earth as you imagine it, and colour in the mantle. Keep the drawing.

2. Which earthquake waves can't cross a liquid? Why?

3. **Your prediction.** Under your feet, is there a sea of lava, or solid rock? What piece of data would decide it?

4. **Your prediction.** Towards the centre of the Earth it gets hotter and hotter. Do you think the hottest part will be liquid or solid? Why?

While you watch the video

1. What two questions can you ask the Earth in order to slice it into layers?

2. Going down with density, what does the Moho jump tell you? And the big jump at about 2,900 km? Why does density increase little by little in the mantle?

3. Compare continental crust and oceanic crust: thickness, density, rocks and age.

4. Why is the Moho deeper under the Pyrenees than under the west of the Iberian Peninsula? (Answer this when you get to Airy's raft.)

5. In the Fiji record section, how far does the S wave reach? What does that prove about the mantle?

6. Going down with the S, write down the five zones and how each one behaves.

7. If no S wave reaches the inner core, how do we know it is rigid? What does the small density jump at that boundary tell you?

8. What did Pratt calculate? What did Airy propose? What is isostasy?

9. What did Celsius have carved in 1731? What did he believe? What did Lyell confirm?

10. **Your hypothesis.** How would you tell whether the sea is falling or the land is rising?

11. What pushed Scandinavia down? Why is it still rising, if the ice left about 10,000 years ago? What has to happen underneath?

12. What did Haskell calculate in 1935 from that rise? What is viscosity?

13. How are glacier ice and the mantle alike? Why does pitch fool you the other way round?

14. What happens to the block if you push it for a second? And for 100,000 years?

15. Read $\tau = \eta / \mu$. What happens if the viscosity grows? And if the rigidity grows? What is its value for the mantle?

16. What are vacancies? How does a crystal change shape without melting?

17. **Stop and think.** How can the hottest part be solid?

18. On the graph, what does each line represent? What is the rule for knowing whether something is solid or liquid?

19. Why is the asthenosphere soft, if it isn't molten?

20. Why does rising rock melt? Where was the La Palma magma stored?

21. Why does the melting line suddenly drop at about 2,900 km?

22. What happens at about 5,150 km? Why doesn't the centre melt, if it is so hot?

23. Where does the crossing of the two lines come from? What is not debated?

24. With the two cuts on one radius, where does the Moho fall? What is the lithosphere?

25. What moves the mantle? What generates the magnetic field?

26. Why don't the mass and the moment of inertia prove the model on their own? What is it that gives confidence?

In class: Haskell's sum, with a calculator

In 1935, Norman Haskell used the rise of Scandinavia to calculate the viscosity of the mantle. The idea is simple: a more viscous mantle takes longer to rearrange itself. You are going to try three viscosities and see which one fits how much the land has risen.

The data.

- Mantle density: $\rho = 3{,}300$ kg/m^3. Gravity: $g = 9.8$ m/s^2. One year: 3.16×10^7 s.

- Half-width of the ice sheet: L = 700 km = 7×10^5 m.

- What was still to rise when the ice left, about 9,700 years ago: $w_0 = 320$ m.

The formulas.

- $k = \pi / (2 \cdot L)$

- The time it takes to rearrange itself: $\tau = 2 \cdot \eta \cdot k / (\rho \cdot g)$, in seconds. Convert it to years.

- What is still to rise after a time t: $w = w_0 \cdot e^{(-t/\tau)}$

- What has risen: $w_0 - w$

- The rate of rise today: w / τ

1. Calculate

Complete the table. The first row, as an example, already has the time.

η (Pa·s)	τ (years)	Still to rise today (m)	Has risen (m)	Rate today (mm/year)
10^{20}	440			
10^{21}				
10^{22}				

2. Compare

What was measured: about 285 m on Sweden's High Coast, almost 10 mm a year in the Gulf of Bothnia, and a few tens of metres still to rise.

1. Which viscosity fits best? Why don't the other two work?

2. What would happen to today's rise if the mantle were a hundred times less viscous?

3. The figure of 320 m comes from a model with that same viscosity. Which part of your result is evidence, and which part isn't?

4. Write down two things this model leaves out.

3. If there is time: the two lines

On graph paper, with depth on the horizontal axis and temperature on the vertical axis, draw the two lines. In red, the real temperature; in black, the melting temperature. Shade what is solid in blue and what is liquid in orange. They are estimates: below 660 km, bear in mind that the margin is large.

Depth (km)	Real temperature (°C)	Melting temperature (°C)
0	15	1,120
100	1,350	1,520
660	1,700	2,165
1,500	1,960	2,900
2,500	2,240	3,600
2,891, above the edge	3,770	3,900
2,891, below the edge	3,770	3,490
4,000	4,660	4,480

Depth (km)	Real temperature (°C)	Melting temperature (°C)
5,150	5,230	5,230
6,371	5,330	5,540

Practice

1. Say which layer you are in, with the two cuts, at 20 km under a continent, at 150 km under an ocean, at 3,000 km and at 6,000 km.

2. At a certain depth, the real temperature is 4,500 °C and the melting temperature is 4,300 °C. Say whether it is solid or liquid, which layer you could be in and what it would be if the two figures were swapped.

3. Explain in your own words how the mantle can let S waves through, and still flow.

Keep thinking

Two different cuts of the same Earth, and both are correct. What does it mean for a model to be true?

And go back to your drawing from the start. What would you change?

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